Fiscal Year:	FY 2024	**Task Last Updated:**	FY 12/20/2024
PI Name:	Hein, Travis Ph.D.		
Project Title:	Effects of Microgravity on Ocular Vascular Hydrodynamics		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Bryan	**State:**	TX
Zip Code:	77807	**Congressional District:**	17
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-HHCHFBP: Human Health Countermeasures, Human Factors, Behavioral Performance. Appendix D
Start Date:	12/21/2018	**End Date:**	07/14/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
Contact Email:	janice.zawaski@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 07/14/2025 per NSSC information (Ed., 7/23/24).		
Key Personnel Changes/Previous PI:	Per the PI, three Co-investigators were added in the fall of 2023: Walter Cromer, Shannon Glaser, and Sanjukta Chakraborty. All are with the Texas A&M University System (Ed., 1/23/25). As the previous Principal Investigator (David Zawieja, Ph.D.) is retiring from Texas A&M University, the new PI for this project is Travis Hein, Ph.D. (also at Texas A&M University). Dr. Zawieja remains as a CoInvestigator on the project.		
COI Name (Institution):	Loerch, Linda (NASA Johnson Space Center) Tharakan, Binu (Scott & White Memorial Hospital) Macias, Brandon (NASA Johnson Space Center) Lee, Stuart (Wyle Laboratories, Inc./NASA Johnson Space Center) Zawieja, David (Texas A&M University System) Bagher, Pooneh (Texas A&M University System) Cromer, Walter (Texas A&M University System) Glaser, Shannon (Texas A&M University System) Chakraborty, Sanjukta (Texas A&M University System)		
Grant/Contract No.:	80NSSC19K0392		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	Spaceflight-associated neuro-ocular syndrome (SANS) is reported to affect ~40% of astronauts completing long-duration spaceflights (as of May 2017) and has been characterized as the development of one or more findings: optic disc edema, hyperopic shifts, globe flattening, cotton-wool spots, or choroidal folds. The leading hypothesis for the development of ocular changes is that prolonged exposure to the headward fluid shift that occurs in weightlessness is the primary instigating factor, and additional factors such as genetic disposition, ambient CO2 on the International Space Station, or on-orbit exercise countermeasures may augment or diminish the development of ocular symptoms. However, the pathophysiology of SANS remains unclear. Evidence for the contribution of intracranial pressure alone in SANS is controversial. Therefore, studies of ocular vascular hydrodynamics are required to clarify if chronic mild elevations of ocular pressure variables compromise ocular structure and function. Since all blood and lymph vessels are compliant, fluid-filled structures whose pressures are strongly influenced by gravity, we propose to focus our studies on the potential changes directly to the ocular vasculature caused by microgravity. Perfusion of the optic nerve and inner retina for sufficient delivery of oxygen and nutrients is dependent on retinal blood flow. The pressure gradient for driving blood flow through the inner retina begins with the arterial pressure in the feed artery, which is the central retinal artery in humans. Changes in retinal blood flow or pressure may contribute to the formation of cotton wool spots and optic disc edema. Optic disc edema, choroidal folds, and optic nerve thickening may also result from ocular venous congestion and/or elevated venous compliance, disruption of the blood-retinal barrier, and/or reduction in ocular lymph flow. There has been no systematic analysis of the ocular vascular changes in microgravity. We have assembled a team of experts in SANS and all 3 main vascular types (arteries, veins, and lymphatics) to address this information gap. Thus, the objective of this application is to determine whether microgravity alters the structure and function of the ocular vasculature at the level of feed arteries, venous exchange and capacitance vessels, and lymph vessels. This provides a novel comprehensive evaluation of the ocular vascular elements. The central hypothesis of this proposal is that microgravity/spaceflight-induced changes in the structure/function of the ocular vasculature lead to alterations in ocular hydrodynamics and promote symptoms of SANS. We will accomplish this objective using in vivo measures of vascular function (retinal artery blood flow, retinal arteriole and venular diameter measurements, and retinal venular permeability measures) and in vitro studies of freshly isolated vascular structure and function (vessel/tissue histology, arterial vasomotor regulation, venous compliance measures, and lymphatic transport characteristics). These studies will be conducted in mice flown in space and the corresponding ground controls to address the following specific aims: 1: Evaluate the effects of microgravity on ocular artery structure/function. 2: Evaluate the effects of microgravity on ocular vein structure/function. 3: Evaluate the effects of microgravity on ocular lymphatic structure/function. Information from these novel studies will provide the first comprehensive analysis of the effects of microgravity on ocular vascular function where the predominant changes associated with SANS in astronauts occur. It will also help define the roles these may play in the etiology of SANS and could lead to the development of countermeasures for SANS.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The expected findings from the experimental planned for RR23 will also provide new insight into vascular complications relevant to ocular diseases in humans on Earth, such as glaucoma, diabetic macular edema, and ocular hypertension.
Task Progress:	We have performed experiments to supplement our ocular vascular (artery and lymphatics) findings in mice after spaceflight on the International Space Station (ISS) in the RR-23 mission on the ISS. We observed decreased ocular artery contractile function and ocular lymphatic phasic activity in the RR-23 mice, which may result from microgravity and/or ionizing radiation. Therefore, the objective of this additional work is to determine whether ionizing radiation exposure in pigs alters the ocular contractile function and gene expression of ocular arteries/retinal tissue and cervical lymphatics. Because we found significant changes in pathology and mRNA expression in livers from RR-23 mice, pig liver morphology and gene expression were also assessed after radiation exposure. We obtained eyes, lymph nodes, and liver tissue from pigs 4 weeks after exposure to various amounts of X-ray radiation. So far, the mRNA expression has been measured in the retinal and liver tissues of the irradiated pigs. We have been able to perform expression profiling of 91 gene transcripts that are known to be altered in the vasculature during disease progression and also show differential response to radiation, aging, and oxidative stress. These pathways include endothelial cell biology, hypoxia signaling pathway, cellular senescence, aging, oxidative stress, inflammatory cytokines, mitochondria, nitric oxide signaling pathway and mitogen-activated protein kinase family signaling. With the limited sample size (1 control and 1-2 per radiation group) so far, we are not able to perform statistical analysis, but changes are evident in our preliminary results with several increases in mRNA expression at the two highest doses of radiation and decreases in several transcripts at the lowest dose of radiation for the retinal tissue samples. Although we have only assessed one liver sample from each of 3 different radiation treatment groups, the results are interesting with most changes showing a decrease in gene expression at a lower radiation dose, but most of the genes (76 out of 91) increased in the liver samples exposed to the highest radiation dose. Future studies are planned to increase the sample size for the radiation treatment groups to allow for sufficient gene analysis and insight into the impact of radiation on gene expression in the retina and liver. Experiments are currently being performed to assess the gene expression changes in the lymph node samples from the irradiated pigs.
Bibliography Type:	Description: (Last Updated: 07/07/2026)
Abstracts for Journals and Proceedings	Zawieja DC, Cromer WE, Bagher P, Tharakan B, Hein TW. "Effects of spaceflight on mouse ocular vascular hydrodynamics and implications for SANS development." 2024 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 13-16, 2024. , Feb-2024

Abstracts for Journals and Proceedings	Muthusamy S, Hunter F, Waithe O, Laurie SS, Macias BR, Bagher P, Hein TW, Zawieja DC, Tharakan B. "Evidence of blood-retinal barrier hyperpermeability in mice exposed to microgravity." 2024 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 13-16, 2024. , Feb-2024